

Lab 3: Linux Commands - II

Write the following Scripts and Execute in Linux Terminal

1. Write a shell script to print your name.

```
read -p "Enter your Name: " PERSON
echo "Hello! $PERSON"
```

```
harsh@Ubuntu:~/Desktop/lab3$ sh hello.sh
Enter your Name: Harsh
Hello! Harsh
```

2. Write a shell script to find whether a number is even or odd.

```
read -p "Hello user, Enter a number to check whether it is odd or even: " num
if [ "$(expr $num % 2)" -eq 0 ]
then
    echo "$num is even"
else
    echo "$num is odd"
fi
```

```
harsh@Ubuntu:~/Desktop/lab3$ sh evenodd.sh
Hello user, Enter a number to check whether it is odd or even: 55
55 is odd
harsh@Ubuntu:~/Desktop/lab3$ sh evenodd.sh
Hello user, Enter a number to check whether it is odd or even: 44
44 is even
```

3. Write a script to print a table of a given number.

```
echo "--- Enter Number to Generate Multiplication Table ---"
read -p "Enter the number : " num
i=1
while [ $i -le 10 ]
do
    echo " $num * $i = `expr $num \* $i` "
    i=`expr $i + 1`
done
```

```
harsh@Ubuntu:~/Desktop/lab3$ sh table.sh
--- Enter Number to Generate Multiplication Table ---
Enter the number : 13
13 * 1 = 13
13 * 2 = 26
13 * 3 = 39
13 * 4 = 52
13 * 5 = 65
13 * 6 = 78
13 * 7 = 91
13 * 8 = 104
13 * 9 = 117
13 * 10 = 130
```

4. Write a shell script to check whether a given no. is prime or not.

```
read -p "Enter a number: " num
i=2
f=0
while [ $i -le `expr $num / 2` ]
```

```

do
if [ `expr $num % $i` -eq 0 ]
then
f=1
fi
i=`expr $i + 1`
done
if [ $f -eq 1 ]
then
echo "$num is composite (not prime)"
else
echo "$num is Prime"
fi

```

```

harsh@Ubuntu:~/Desktop/lab3$ sh prime.sh
Enter a number: 12
12 is composite (not prime)
harsh@Ubuntu:~/Desktop/lab3$ sh prime.sh
Enter a number: 13
13 is Prime

```

5. Write a shell script to find the simple interest.

```

read -p "Enter the principle value: " p
read -p "Enter the rate of interest: " r
read -p "Enter the time period: " t
i=`expr $p \* $t \* $r`
j=`expr $i / 100`
echo "Simple Interest is: $j"

```

```

harsh@Ubuntu:~/Desktop/lab3$ sh simpleinterest.sh
Enter the principle value: 20000
Enter the rate of interest: 5
Enter the time period: 2
Simple Interest is: 2000

```

6. Write a shell script to find sum of 'n' numbers.

```

read -p "How many numbers do you want to sum up : " N
i=1
sum=0
while [ $i -le $N ]
do
read -p "Enter number $i : " num
sum=$((sum + num))
i=$((i + 1))
done
echo "Sum : " $sum

```

```

harsh@Ubuntu:~/Desktop/lab3$ sh nsum.sh
How many numbers do you want to sum up : 6
Enter number 1 : 8
Enter number 2 : 6
Enter number 3 : 9
Enter number 4 : 4
Enter number 5 : 1
Enter number 6 : 3
Sum : 31

```

7. Write a shell script to find the largest number of three numbers.

```
read -p "Enter Num1 : " num1
read -p "Enter Num2 : " num2
read -p "Enter Num3 : " num3

if [ $num1 -gt $num2 ] && [ $num1 -gt $num3 ]
then
    echo "The largest numbers among $num1, $num2, $num3 is : "$num1
elif [ $num2 -gt $num1 ] && [ $num2 -gt $num3 ]
then
    echo "The largest numbers among $num1, $num2, $num3 is : "$num2
else
    echo "The largest numbers among $num1, $num2, $num3 is : "$num3
fi
```

```
harsh@Ubuntu:~/Desktop/lab3$ sh largest3.sh
Enter Num1 : 6
Enter Num2 : 8
Enter Num3 : 4
The largest numbers among 6, 8, 4 is : 8
```